

Pure Substances and Mixtures — Paper C

HARD • NON-CALCULATOR • 45 MINUTES • 30 MARKS

NAME

DATE

SCORE / 30

- Answer **every** question in the space provided.
- No calculator is needed anywhere on this paper.
- Where a question says **explain**, your answer must say whether the atoms are **chemically joined** or **not joined**, and whether there is **one kind of particle** or more than one.
- The mark for each question is shown in square brackets on the right.
- Several questions here describe substances or experiments you may not have seen before. You are not expected to recognise them — apply the ideas you already have.
- "Pure" on this paper always has its chemical meaning: one substance only.

SECTION 1 – HEATING CURVES

- 1.** Two solid samples are heated at the same steady rate and their temperatures recorded every minute.

Time (min):	0	1	2	3	4	5	6	7
Sample X:	40	60	80	80	80	80	95	110
Sample Y:	40	60	71	74	77	79	94	109

- (a)** State the melting point of sample X, and how you can tell it is pure. [2]

- (b)** State the temperature range over which sample Y melts. [1]

- (c)** Explain what the readings for Y tell you about the sample, giving two reasons. [3]

2. Explain, in terms of particles, why an impurity makes a solid start to melt at a lower temperature than the pure solid. [3]

SECTION 2 – PLANNING A TEST

3. A student is given a white powder and told it is either pure sodium chloride or a mixture of sodium chloride and sand. She has a beaker, water, filter paper and a funnel. Describe a test she could do, the result that would show it is a mixture, and the result that would show it is pure. [4]

4. Salt is spread on icy roads in winter.

- (a) Explain, using the idea of impure substances, why the ice melts even though the air is below 0 °C. [2]

- (b) A student says the salt "heats up the ice". Explain what is wrong with this, and what has actually changed. [2]

SECTION 3 – UNFAMILIAR SUBSTANCES

5. Each description below is of an unfamiliar substance. For each, state whether it is an **element**, a **compound** or a **mixture**, and give a reason.

- (a) Neon: every particle is a single atom, and every atom is the same. [2]

(b) Ammonia: every particle is one nitrogen atom joined to three hydrogen atoms. [2]

(c) Brass: copper atoms and zinc atoms packed together in a solid, in any proportion the maker chooses, not joined to each other. [2]

6. A student says: "Carbon dioxide must be a mixture, because it is made from carbon and oxygen and both of those are elements." Explain why the student is wrong. [3]

7. Sea water boils at a slightly higher temperature than pure water, and the temperature keeps rising as it boils. Explain both observations. [4]

End of paper. Check every answer has a unit or a form the question actually asked for.